

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

CON332

FinOps adopting Kubernetes: Optimizing cost for business innovation

Sandhya Job

(she/her)

WW Senior Container Specialist, EKS
Amazon Web Services

Rachel Leekin

(she/her)

Container Specialist Solutions Architect, FSI
Amazon Web Services



Agenda

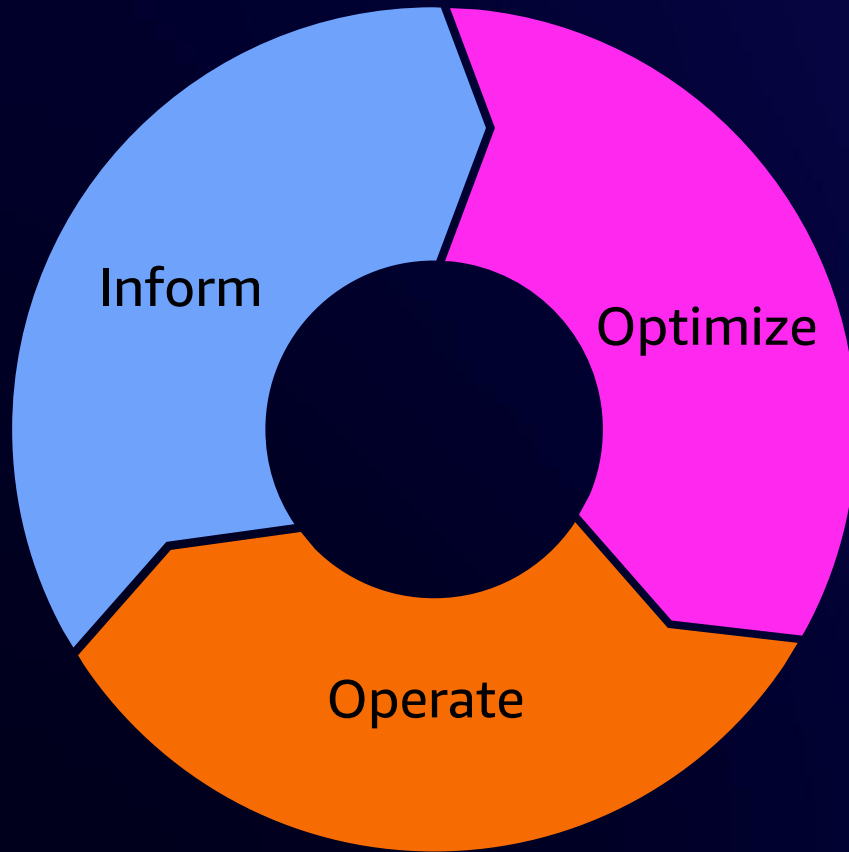
- 01 Your speakers
- 02 What is FinOps?
- 03 Cost visibility in Kubernetes shared environment
- 04 Strategies to optimize Amazon EKS deployment
- 05 Inform, optimize, operate!
- 06 Takeaways

Tenets for cost management



FinOps and phases

FINOPS: A CULTURAL PRACTICE ENABLING DATA-DRIVEN SPENDING DECISIONS



1. INFORM

Visibility of allocation and shared accountability

2. OPTIMIZE

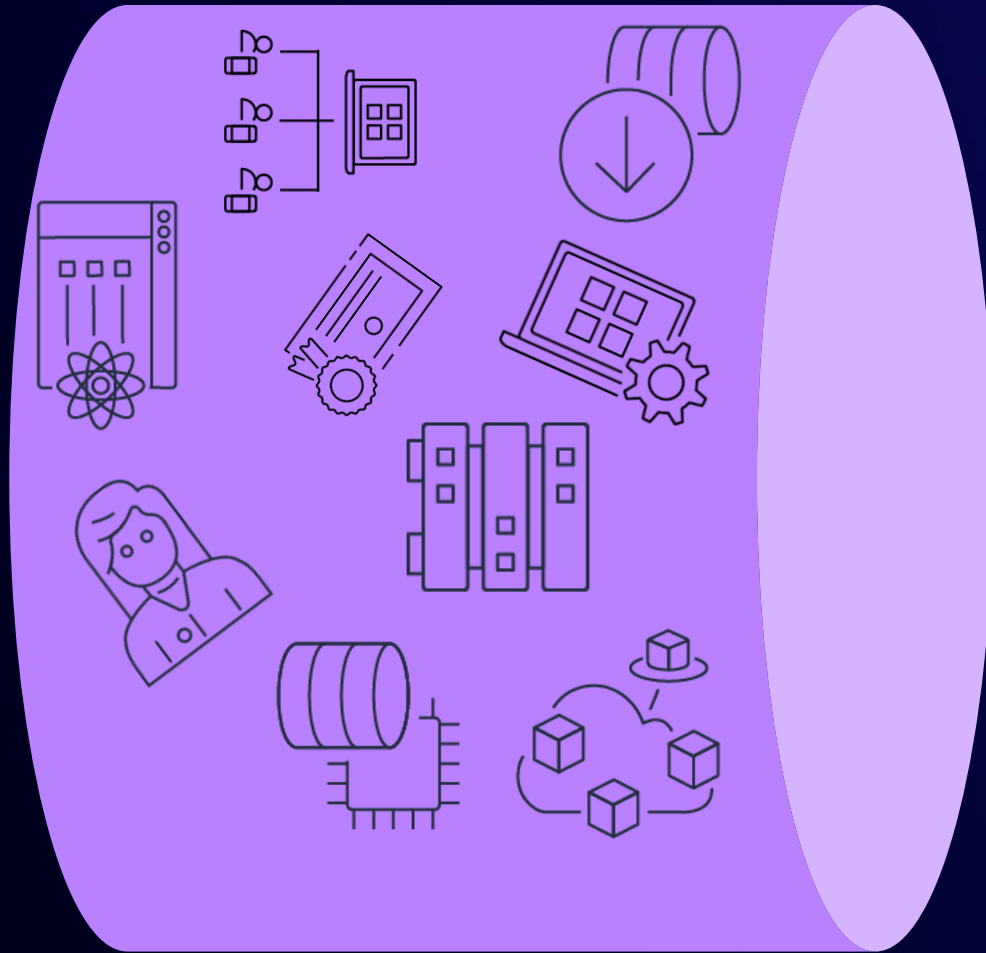
Continual improvement in efficiency

3. OPERATE

Implementation of processes that drive business goals

Source: FinOps Foundation, FinOps Framework, finops.org/introduction/what-is-finops/

Cloud usage and cost



Infrastructure



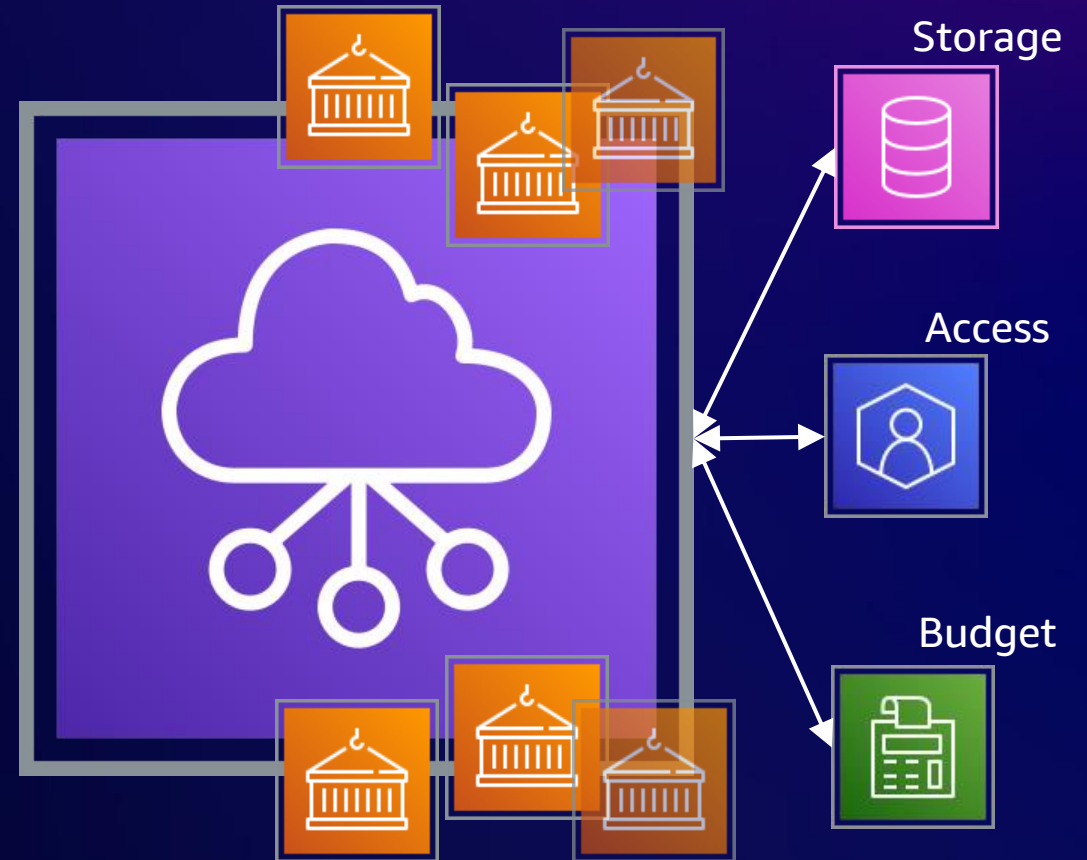
Data



Resources

Modern application features

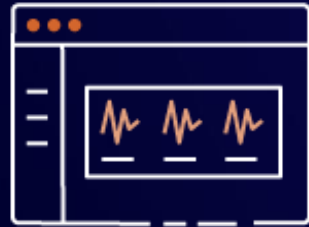
- Highly scalable
- Variable workloads
- Shared platform and resources
- Shared responsibility model



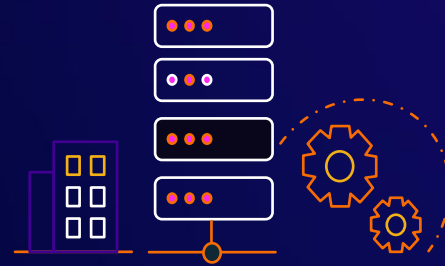
Amazon EKS cost optimization strategies



Observability



Scaling and optimization



Hardware



Key performance indicators (KPI) and automation



Purchase options

Inform

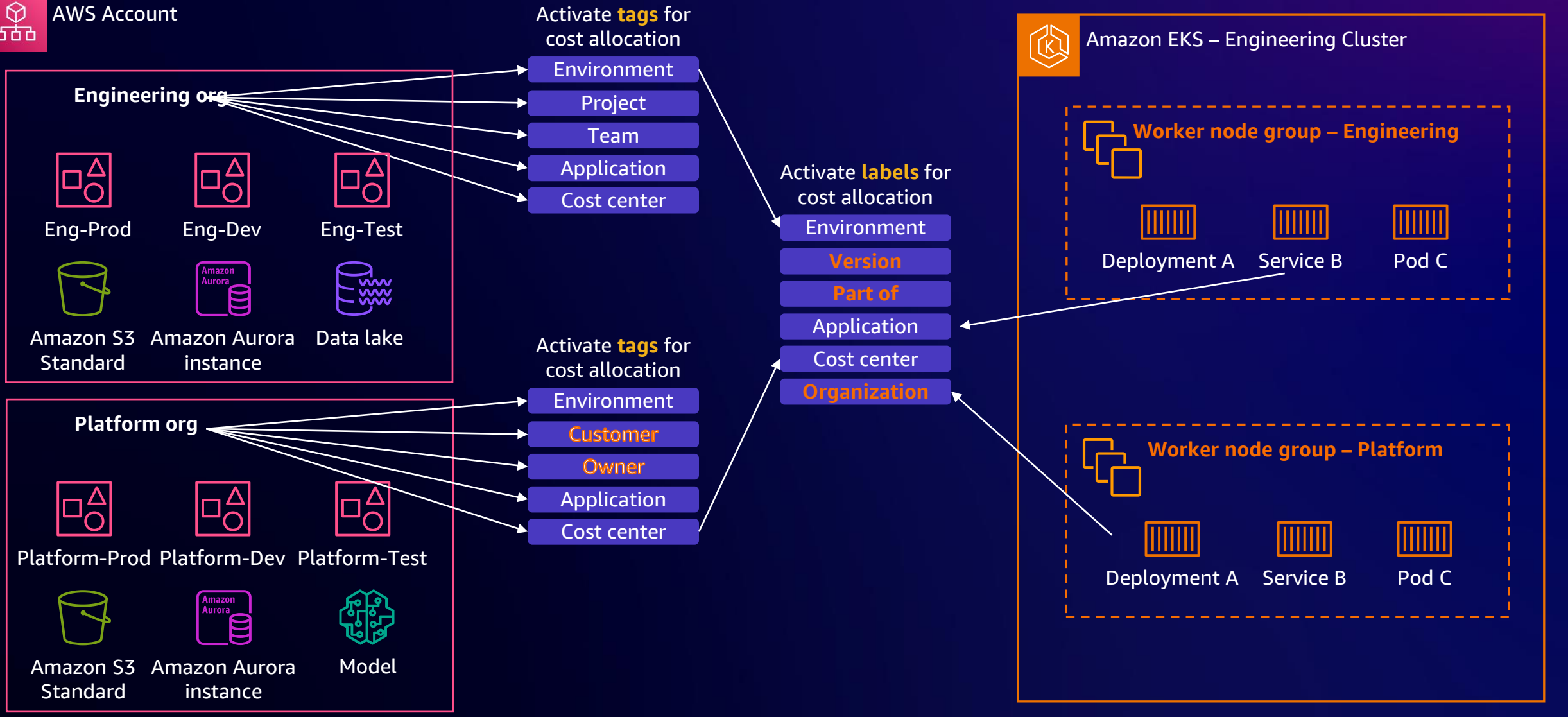
Optimize

Operate

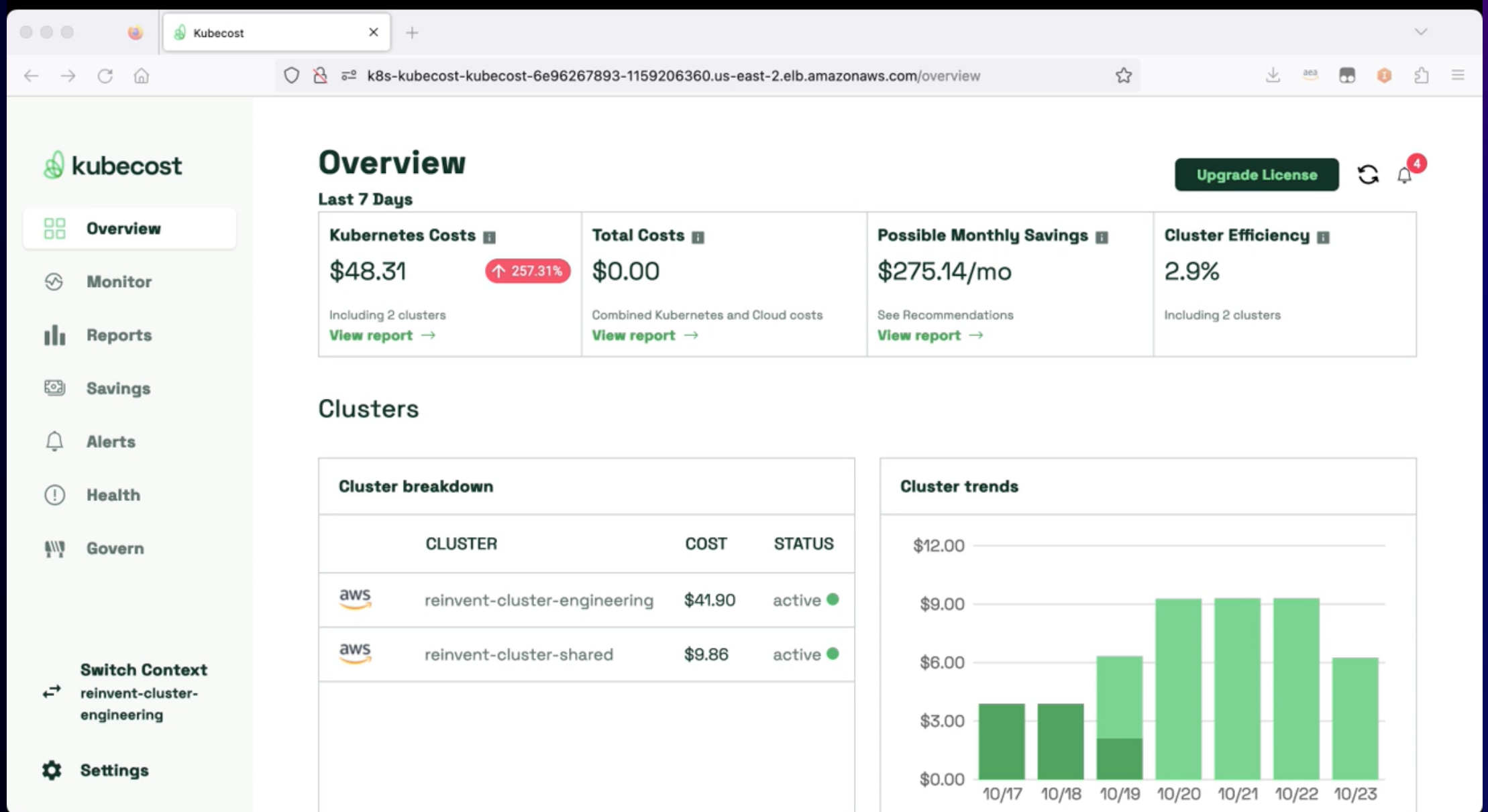
Inform: Tagging and labels in layers



AWS Account



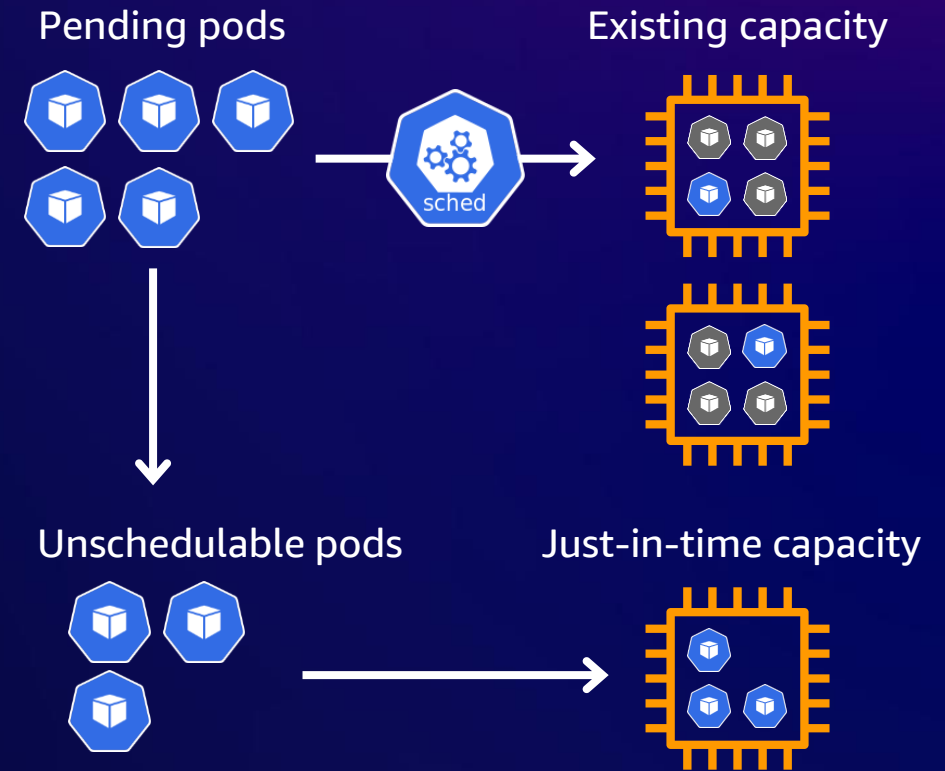
Inform: KubeCost demo



Optimize: Karpenter and flexible compute

Open source, flexible, and high-performance Kubernetes cluster auto scaler

- Supported by AWS in Amazon EKS clusters
- Work in tandem with the Kubernetes scheduler and Amazon EC2
- Launch right-sized nodes in response to unschedulable pods without Amazon EC2 Auto Scaling groups
- Natively work with Kubernetes scheduling constraints, such as node label selectors, pod affinity, and topology spread constraints
- Terminates unused nodes and consolidates pods



Optimize: Karpenter whiteboarding

Operate: Integration with business operations

- Defined financial responsibilities
- KPIs on thresholds
- Automated budgeting and alerts
- Product-level cost management
- Regular finance forecasting



Operate: FinOps operations whiteboarding

Takeaways

- Integrated FinOps process with business operations
- Optimized modern architecture
- Incorporated cost-saving mechanisms
- Kubernetes evolves your FinOps process

Learn more

https://aws.github.io/aws-eks-best-practices/cost_optimization/cfm_framework/



AWS joined the FinOps Foundation in October 2023



Thank you!



Please complete the session survey in the mobile app

Sandhya Job

sandyjob@amazon.com

LinkedIn: @sandhya-job

Rachel Leekin

rleekin@amazon.com

LinkedIn: @rachel-leekin

